

# Assessing the Potential Impact of Real-time Crowding Information on Metro Manila's Train Lines

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**Abstract**—This study explores the potential impact of Real-Time Crowding Information (RTCI) on commuter behavior in Metro Manila's train systems (LRT-1, LRT-2, and MRT-3). Overcrowding in trains remains a significant issue, causing stress and discomfort, particularly during peak hours. This study aims to determine whether access to real-time crowding information could influence commuters to adjust their regular travel behavior. A mixed-methods approach was employed, using surveys to assess crowding experiences, commuter satisfaction, and behavioral responses to RTCI. The findings suggest that RTCI has the potential to alleviate discomfort by enabling passengers to make more informed travel decisions. However, awareness of RTCI alone did not strongly influence commuter behavior, indicating that factors such as trust in the accuracy of the data and the inclusion of more tangible benefits are crucial. Ultimately, this study highlights the practical implications of RTCI, suggesting that its integration into Metro Manila's train systems could improve commuter experiences by reducing overcrowding and enabling more efficient travel decisions.

**Index Terms**—Commuter Behavior, Overcrowding, Real-Time Crowding Information (RTCI), Metro Manila Train Lines, Public Transportation

## I. INTRODUCTION

Public transportation remains a lifeline for millions of commuters in Metro Manila, with train systems such as LRT-1, LRT-2, and MRT-3 playing a critical role in daily commuting. However, these train lines have become a significant source of frustration and discomfort due to notorious overcrowding, as they struggle to meet the ever-growing demand for efficient public transportation [1]. Despite their importance, the system's infrastructure has failed to keep pace with the city's rapid growth, leaving existing facilities insufficient to accommodate the rising commuter demand.

This study aims to explore train commuters' experiences and perceptions of overcrowding on Metro Manila's train lines and investigate how they might respond to the introduction of Real-Time Crowding Information (RTCI). Specifically, it examines whether access to real-time crowding data would influence commuters' decisions to seek alternative transportation or adjust their travel times.

RTCI systems in international contexts have demonstrated benefits in influencing commuter behavior. For instance, a study in Stockholm showed how RTCI helped redistribute

passengers across train cars, reducing congestion in busy sections [2]. Similarly, a simulation in Kraków indicated that RTCI could alleviate severe overcrowding by up to 30%, improving travel satisfaction during peak hours [3]. In line with these findings, a study on bus bunching found that providing RTCI can influence bus passengers to intentionally skip an overcrowded bus and wait for a later departure [4]. While RTCI systems have demonstrated significant benefits in influencing commuter behavior in other global cities, there is limited research on its potential effectiveness in Metro Manila, where commuter behavior and transportation infrastructure differ from other urban settings.

Prior studies have consistently shown that overcrowding leads to increased discomfort, longer waiting times, and higher stress levels among passengers, all of which negatively affect the commuting experience [5] [6]. Recent research highlights RTCI as a promising solution to these challenges, with studies from around the world indicating that providing real-time crowding information can help passengers make more informed decisions and adjust their routes or travel times to avoid congestion, thus enhancing their overall travel satisfaction [7].

A study on crowd dynamics found that simulating commuter movement is key to designing effective crowd control measures and layouts to reduce congestion [8]. In this context, understanding crowd behavior is also crucial. Modern scholars emphasize the need for a nuanced understanding of how perceived crowding influences travel decisions. Research indicates that commuters often alter their routes or travel times to avoid congested conditions [9], yet the effectiveness of RTCI in mitigating these effects has not been thoroughly validated in the context of Metro Manila. While the literature provides a strong foundation for understanding RTCI's role in public transport and its benefits in redistributing passengers and alleviating congestion, there remains a significant gap in empirical research addressing its application to Metro Manila's train systems. This study aims to bridge the gap by examining the potential impact of RTCI on commuter travel decision-making in Metro Manila and contributing valuable insights into its ability to alleviate overcrowding in this unique context.

### Statement of the Problem

The issue of overcrowding in Metro Manila's train systems has become increasingly urgent, particularly with the opening of the first phase of the LRT-1 Cavite Extension. This expansion is projected to serve an additional 20,000 to 30,000 passengers daily, exacerbating congestion on trains and platforms [10]. The issue was highlighted by a recent interview with Enrico Banipayo, the CEO of Light Rail Manila Corporation (LRMC), who has already seen an unexpected increase of around 50,000 additional riders in just a few days, illustrating the immediate strain on the system as demand surges [11]. As the demand for public transportation continues to rise, the existing infrastructure is likely to struggle further, compromising passenger comfort and safety while contributing to longer wait times. Existing literature highlights the psychological toll of overcrowding, including heightened stress and dissatisfaction, which can adversely impact commuter productivity and overall well-being [5] [6].

Despite the growing demand for efficient public transportation, especially during peak hours, Metro Manila's train systems remain ill-equipped to meet commuter needs. The lack of effective crowd management strategies and real-time information systems further exacerbates these issues.

To guide this study, the following research questions and hypotheses are addressed:

### Research Questions:

1. What are commuters' overall experiences with crowding and train waiting times when using Metro Manila's train lines (LRT-1, LRT-2, and MRT-3)?
2. Would access to Real-Time Crowding Information (RTCI) impact commuters' decisions to travel on Metro Manila's train lines?

### Hypotheses:

1. Commuters who use Metro Manila's train lines (LRT-1, LRT-2, and MRT-3) experience longer waiting times and higher levels of discomfort due to platform overcrowding.
2. Access to Real-Time Crowding Information (RTCI) will significantly influence commuters' decisions to travel on Metro Manila's train lines, leading them to adjust their travel times or seek alternative modes of transportation.

### Scope and Delimitation

This study focused on the experiences and perceptions of train commuters regarding platform overcrowding in Metro Manila, specifically across LRT-1, LRT-2, and MRT-3. It examined how commuters cope with overcrowding and explored their potential responses if RTCI were introduced.

### Scope:

- **Geographic Scope:** The study focused on commuters using Metro Manila train lines (LRT-1, LRT-2, and MRT-3).
- **Demographic Scope:** Data was collected from a variety of train commuters, with no exclusions in demographics.
- **Conceptual Scope:** The study explored key concepts such as RTCI, platform overcrowding,

commuter behavior, commuter comfort, and overall commuter travel experience.

### Delimitations:

- **Geographic Limitation:** The study was limited to the LRT-1, LRT-2, and MRT-3 train lines and excluded other forms of public transportation within Metro Manila.
- **Technological Limitation:** The study assumed the hypothetical availability of RTCI but did not investigate its technicalities or infrastructure required for its implementation.
- **Temporal Limitation:** Data was collected during a limited time period, which may not have fully captured variations in commuter behavior.
- **Sampling Limitation:** Convenience sampling was used to gather respondents, chosen based on accessibility rather than random selection. The findings may not have fully represented the broader commuter population of Metro Manila's train lines.
- **Exclusions of Additional Factors:** The study did not address factors such as fare prices, security issues, or infrastructure concerns that may also influence commuters' travel decisions.

### Significance of the Study

The findings of this study contribute to addressing the issue of overcrowding in Metro Manila's train systems. Existing research has found that RTCI has the potential to positively influence commuter behavior, support better passenger distribution, and reduce crowding in certain situations.

Furthermore, the findings could provide essential data for developing future RTCI applications and guide transit authorities' decisions in integrating crowd-management technologies into the current public transport system. This study also serves as a basis for further studies on RTCI and how it can improve public transportation and contribute to better urban transport management.

## II. METHOD

### A. Research Design

This study used a Descriptive-Correlational Research Design in collecting and analyzing data on commuter experiences with overcrowding and responses to the potential implementation of RTCI across Metro Manila's main train lines. This approach allowed us to gather and analyze data that helped in understanding current commuter challenges and examining relationships between factors related to commuting, such as crowding, waiting times, and commuting frequency. We explored how they relate to commuters' likelihood of using RTCI to adjust their commuting behavior.

### Quantitative Data Collection

To address the study's research questions, quantitative data were collected through a structured survey created for commuters across Metro Manila's train lines. The survey included closed-ended questions and Likert scale items to measure key variables such as comfort level, experiences with crowding, and likelihood to use RTCI. This quantitative approach enabled statistical analysis of trends in commuter behavior and perceptions of overcrowding.

### *Qualitative Data Collection*

For the qualitative component of this study, the survey included open-ended questions to capture detailed responses on topics related to RTCI and potential improvements to commuters' travel experiences. The qualitative responses provided context and depth to the quantitative findings. It enabled the analysis of commuter perspectives not possible using quantitative methods.

#### *B. Population and Sample*

The target population for this study included a combined daily commuter population of approximately 829,529 individuals across Metro Manila's three main train lines [12] [13] [14].

- 323,000 daily commuters who use the LRT-1
- 149,331 daily commuters who use the LRT-2
- 357,198 daily commuters who use the MRT-3

Given the practical constraints in collecting data from the total commuter population across Metro Manila's train lines, a non-probability sampling method, specifically convenience sampling, was chosen for this study. This method enabled the collection of responses from a diverse yet accessible subset of commuters. Links to the survey were distributed through social media and through approaching individuals directly at one selected train station on a specific day.

#### *Sample Size*

The survey resulted in a final sample size of 252 respondents. This smaller sample and sampling method is recognized as a limitation in this study and it may affect the generalizability of the results to the entire train commuter population of Metro Manila. As such, the conclusions drawn should be viewed as preliminary and indicative rather than definitive.

#### *C. Respondent/Participant*

The study's respondents are regular commuters on Metro Manila's LRT-1, LRT-2, and MRT-3 train lines. Using convenience sampling, respondents were selected based on accessibility and willingness to participate. Each respondent completed a structured survey that included both closed-ended questions, to assess variables like commuter discomfort and waiting times, and open-ended questions, to gather personal insights into their commuting experiences and provide a deeper understanding of the topic.

#### *D. Instruments*

To collect data for this study, the following instruments were used:

#### *Survey Questionnaire*

A structured survey questionnaire served as the primary tool for collecting quantitative data from commuters on the LRT-1, LRT-2, and MRT-3 lines. The questionnaire included closed-ended questions and Likert scales designed to measure key factors such as:

- Commuter Discomfort: Measured on a five-point Likert scale from "Very Comfortable" to "Very Uncomfortable."
- Waiting Times: Respondents reported average

waiting times by selecting from defined time intervals.

- Response to RTCI: Measured on a five-point Likert scale to gauge their likelihood of using RTCI to adjust travel behaviors.

In addition to the collecting quantitative data, the survey included open-ended questions on several topics:

- Reasons for choosing to use or not use RTCI
- Specific experiences with overcrowding on Metro Manila's train lines (LRT-1, LRT-2, MRT-3)
- Suggestions for improving the commuting experience
- Additional comments on commuting or the potential implementation of RTCI

These questions were designed to explore the qualitative aspects of commuters' experiences with overcrowding and their perspectives on the potential implementation of RTCI. Responses were systematically coded and analyzed to identify recurring themes in their experiences with Metro Manila's public train systems and their sentiments toward RTCI. This qualitative analysis provided context and depth to the quantitative findings, complementing the statistical trends observed.

#### *E. Data Gathering Procedure*

This section outlines the data collection methods and processes used in the study on overcrowding across Metro Manila's LRT-1, LRT-2, and MRT-3 train lines, as well as the potential impact of RTCI on commuter behavior.

#### *Sources of Data*

Data were collected from primary sources through a structured survey. This survey was designed to gather firsthand information directly from commuters on the LRT-1, LRT-2, and MRT-3.

#### *Data Collection Methods*

A structured survey questionnaire was used as the primary data collection method. This survey was administered both in-person and online. This allowed for a flexible and accessible approach to reaching commuters on Metro Manila's LRT-1, LRT-2, and MRT-3 train lines. Online distribution occurred via social media and social messaging applications, while in-person distribution used posters and printed QR codes distributed at a train station.

#### *Step-by-Step Data Collection Process*

- Recruitment of Respondents: Using convenience sampling, we reached out to Metro Manila train commuters online and approached commuters directly at the Ayala MRT-3 station on November 5, 2024.
- Survey Administration: The structured survey link was shared via social media and social messaging applications. Posters and printed handouts with QR codes and Google Form links were physically presented to potential respondents at a train station. This allowed commuters to easily access the survey on their mobile devices while minimizing their travel disruption.

- **Data Quality Checks:** Throughout the data collection period of five weeks, responses were periodically reviewed to check for data integrity.

#### *Informed Consent Process*

Participants were informed of the study's purpose, procedures, and their rights, including the right to withdraw at any time without consequence. Consent was obtained via an introductory statement in the online survey. Their confidentiality was assured through the Data Privacy Consent Form.

#### *Data Recording and Storage*

Responses were collected electronically and securely stored, with access limited to authorized research personnel only. To protect participant privacy, all responses were anonymized in compliance with data protection standards.

#### *Ethical Considerations*

The study followed ethical guidelines as set forth by the Philippine Data Privacy Act of 2012 (RA 10173). Participants were assured that their responses would be used responsibly and for research purposes only.

#### *Time Frame of Data Collection*

Data collection occurred over a specified period of five weeks, from October 14, 2024, to November 21, 2024.

#### *F. Statistical Treatment of Data and Data Analysis*

This section outlines the methods used to process and analyze the collected survey data to address the research questions and test our hypotheses concerning overcrowding in Metro Manila's train lines and the potential impact of Real-Time Crowding Information (RTCI) on commuter behavior.

#### *Data Preparation:*

- **Data Cleaning:** The raw survey data was checked for errors and inconsistencies. This included identifying duplicate entries and invalid responses.
- **Data Coding:** Responses from the survey were coded to facilitate analysis, such as assigning numerical values to enable statistical calculations.

#### *Statistical Methods:*

- **Descriptive Analysis:** Descriptive analysis was employed to summarize the survey data. This included frequency distributions and measures of central tendency to primarily analyze the respondents' demographic characteristics and assess overall crowding ratings.
- **Correlation Analysis:** Correlation analysis was conducted to examine the relationships between commuter comfort levels and two factors: the frequency of crowding and train waiting times.
- **A Chi-square test and a series of regression analyses** were conducted to test our hypotheses regarding the impact of crowding and waiting time on commuter comfort, as well as the potential influence of RTCI on commuter decision-making.

#### *Statistical Software:*

Microsoft Excel (version 2410) and LibreOffice Calc

(version 24.8.0.3) were used to code the raw survey data to facilitate data analysis. The data was then imported into Jamovi (version 2.3.28), a statistical analysis program, to analyze and visualize the quantitative data collected from the survey.

### III. RESULTS AND DISCUSSION

In this chapter, we present the results of our study using descriptive and regression analysis to explore commuter experiences with overcrowding, waiting times, and comfort levels on Metro Manila's train lines. We also tested our hypotheses to determine the impact of RTCI on commuters' likelihood to adjust their travel plans. For our open-ended survey questions, qualitative analysis provided insights into commuter experiences, challenges, and suggestions regarding their commute on Metro Manila's train lines. This approach allowed us to address our research questions and evaluate the potential of RTCI.

#### *A. Data Presentation and Analysis*

We collected data from a total of 252 respondents through a Google Forms online survey which included demographic questions about the respondents.

Respondent Distribution by Age Group

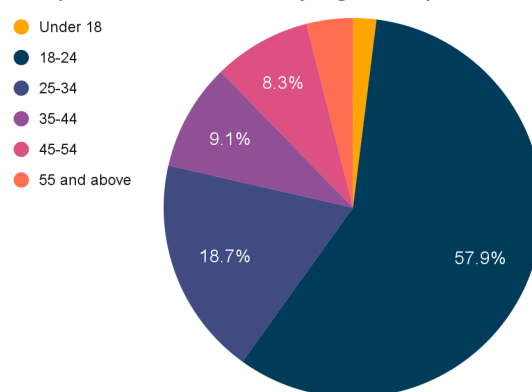


Fig. 1. Pie chart of the distribution of survey respondents by age group

Figure 1 shows the distribution of survey respondents by age group. The majority of respondents (57.9%) fell within the 18-24 age group— young adults likely to include university students and early-career professionals who rely heavily on public transportation and are generally open to new technologies. The second-largest group (18.7%), aged 25–34, represents a key working-age demographic with regular commuting routines. Their responses offered insights into the challenges of commuting during peak-hour congestion.

TABLE I  
CROWDING EXPERIENCE RATINGS FOR PEAK AND NON-PEAK HOURS

| Rating Metric | Non-Peak Hours<br>(9 AM - 4 PM) | Peak Hours<br>(6 AM - 9 AM and<br>4 PM to 7 PM) |
|---------------|---------------------------------|-------------------------------------------------|
| Mean          | 3.52                            | 1.80                                            |
| Median        | 4.00                            | 2.00                                            |
| Mode          | 4.00                            | 1.00                                            |

Table I displays a summary of the measures of central tendency for the crowding experience ratings reported by

survey respondents during peak and non-peak hours. These ratings were measured on a Likert scale from 1 to 5, where 1 represents “Very Poor” and 5 represents “Very Good”. The mean rating for non-peak hours was 3.52, indicating that commuters generally experienced moderate crowding outside of peak times. During peak hours, the mean rating dropped to 1.80, which highlighted increased perceptions of overcrowding. The median values show a similar trend, with non-peak hours rated at 4.00 (indicating a more comfortable experience) and peak hours rated at 2.00 (indicating a less comfortable experience). The mode values further reinforce this pattern, with the most common rating during non-peak hours being 4.00, while peak hours saw a mode of 1.00, suggesting that extreme crowding is frequently experienced during those times. This table highlights a stark contrast in commuter crowding perceptions between peak and non-peak hours.

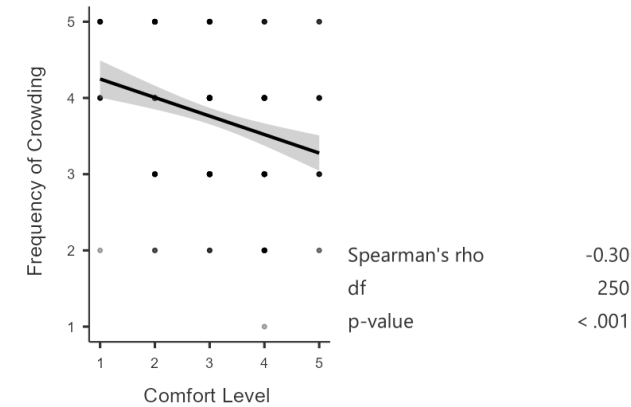


Fig. 2. Relationship between commuter comfort level and train crowding frequency

Figure 2 shows a scatter plot illustrating the relationship between commuter comfort levels and the frequency of encountering overcrowding on Metro Manila’s train lines. The x-axis shows the comfort level (with a Likert scale from “Very Uncomfortable” to “Very Comfortable”) and the y-axis indicates how often they encounter overcrowded platforms (with a Likert scale from “Never” to “Always”). The downward-sloping trend line indicates a negative correlation, suggesting that as the frequency of encountering overcrowded platforms increases, commuter comfort decreases. The shaded area around the regression line represents a 95% confidence interval and the range in which we can expect most of the commuter comfort ratings to fall. With a Spearman’s rho of -0.30, this is a weak negative correlation, but it is still statistically significant with a p-value of <.001. This suggests that the frequency of encountering train overcrowding contributes to reduced comfort, but it is not the only factor influencing it.

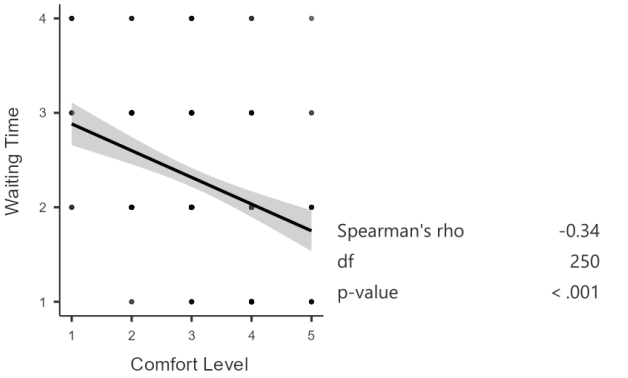


Fig. 3. Relationship between commuter comfort level and average train waiting time

Figure 3 shows a scatter plot illustrating the relationship between commuter comfort levels and average waiting times. The x-axis represents the commuter comfort level while the y-axis indicates waiting time. The responses are coded with "1" representing "Less than 5 minutes", 2 representing "5-10 minutes", 3 representing "11-15 minutes", and "4" representing "More than 15 minutes." The downward-sloping trend line indicates a negative correlation, implying that as waiting times increase, commuter comfort levels decrease. This is a moderate negative correlation (with a Spearman’s rho of -0.34) and is statistically significant (p < .001). This finding suggests that longer waiting times are associated with lower commuter comfort, but waiting time is likely only one of several factors affecting overall comfort.

Hypothesis 1 Testing

“Commuters who use Metro Manila’s train lines (LRT-1, LRT-2, and MRT-3) experience longer waiting times and higher levels of discomfort due to platform overcrowding.”

To test this hypothesis, we conducted a series of regression analyses based on commuting times. Survey respondents were categorized into three groups: non-peak hour commuters, peak hour commuters, and flexible/varied commuters. The overall comfort level was set as the dependent variable, while factors such as waiting time and frequency of crowding were included as covariates. The results provided mixed support for our hypothesis, with differences in their impact.



## Linear Regression

### for Non-Peak Hour Commuters

| Model Fit Measures |      |                |
|--------------------|------|----------------|
| Model              | R    | R <sup>2</sup> |
| 1                  | 0.63 | 0.39           |

| Model Coefficients - Comfort Level |          |      |       |        |
|------------------------------------|----------|------|-------|--------|
| Predictor                          | Estimate | SE   | t     | p      |
| Intercept                          | 2.84     | 0.60 | 4.72  | < .001 |
| Frequency of Crowding              | -0.03    | 0.10 | -0.26 | 0.799  |
| Waiting Time                       | -0.37    | 0.12 | -3.13 | 0.003  |
| Non-Peak Hour Experience           | 0.08     | 0.09 | 0.83  | 0.411  |
| Peak Hour Experience               | 0.50     | 0.10 | 4.82  | < .001 |

Fig. 4. Linear regression results for non-peak hour commuters

Figure 4 shows the first linear regression analysis focused on non-peak hour commuters. It illustrates how comfort levels were influenced by waiting times, crowding frequency, and experiences during peak and non-peak hours. The results indicated a moderate model fit, with an R<sup>2</sup> value of 0.39, suggesting that 39% of the variability in comfort levels could be explained by the predictors. In this group, waiting time had a statistically significant negative effect on comfort (Estimate = -0.37, p = 0.003), indicating that as waiting times increased, comfort levels decreased. Experiences during peak hours positively influenced comfort (Estimate = 0.50, p < .001), indicating that non-peak commuters' perceptions of peak-hour conditions impacted their overall comfort. However, the frequency of crowding and non-peak experiences did not significantly affect comfort. This suggests that, for non-peak commuters, waiting time and perceptions of peak experiences are the more influential factors affecting comfort.

## Linear Regression

### for Peak Hour Commuters

| Model Fit Measures |      |                |
|--------------------|------|----------------|
| Model              | R    | R <sup>2</sup> |
| 1                  | 0.50 | 0.25           |

| Model Coefficients - Comfort Level |          |      |       |        |
|------------------------------------|----------|------|-------|--------|
| Predictor                          | Estimate | SE   | t     | p      |
| Intercept                          | 3.39     | 0.67 | 5.04  | < .001 |
| Frequency of Crowding              | -0.18    | 0.11 | -1.64 | 0.104  |
| Waiting Time                       | -0.37    | 0.11 | -3.25 | 0.002  |
| Non-Peak Hour Experience           | 0.31     | 0.10 | 3.01  | 0.003  |
| Peak Hour Experience               | 0.06     | 0.09 | 0.66  | 0.510  |

Fig. 5. Linear regression results for peak-hour commuters

Figure 5 shows the second regression analysis that evaluated peak-hour commuters. The results yielded a moderate fit R<sup>2</sup> of 0.25. For peak-hour commuters, waiting time was again a significant predictor of comfort level (Estimate = -0.37, p = 0.002). This result reinforces the trend that longer waiting times correlate with lower comfort levels, regardless of peak hour constraints. However, opposite to the non-peak group, comfort was positively influenced by non-peak experiences (Estimate = 0.31, p = 0.003). Peak-hour experiences and crowding frequency did not show a significant relationship with comfort. This finding suggests that peak-hour commuters who also travel during non-peak hours report slightly higher comfort, possibly due to their comparison between crowded and less crowded conditions.

## Linear Regression

### for Flexible Hour Commuters

| Model Fit Measures |      |                |
|--------------------|------|----------------|
| Model              | R    | R <sup>2</sup> |
| 1                  | 0.70 | 0.48           |

| Model Coefficients - Comfort Level |          |      |       |        |
|------------------------------------|----------|------|-------|--------|
| Predictor                          | Estimate | SE   | t     | p      |
| Intercept                          | 1.62     | 0.65 | 2.51  | 0.015  |
| Frequency of Crowding              | -0.07    | 0.13 | -0.52 | 0.604  |
| Waiting Time                       | -0.21    | 0.11 | -1.92 | 0.060  |
| Non-Peak Hour Experience           | 0.51     | 0.09 | 5.49  | < .001 |
| Peak Hour Experience               | 0.21     | 0.12 | 1.71  | 0.093  |

Fig. 6. Linear regression results for flexible/variable hour commuters

Figure 6 shows the third analysis that examined flexible/varied hour commuters. This provided the strongest model fit among the three groups, with an  $R^2$  of 0.48. In this group, comfort levels were significantly, and positively influenced by their non-peak hour experience (Estimate = 0.51,  $p < .001$ ). Their peak hour experience also had a positive effect on comfort, although it was marginally non-significant (Estimate = 0.21,  $p = 0.093$ ). However, neither waiting time nor frequency of crowding showed a significant effect on comfort levels ( $p = 0.060$  and  $p = 0.604$ ). These results suggest that flexible commuters' comfort levels are influenced more by the quality of their experiences at various times, rather than by the logistical challenges of waiting and overcrowding. This pattern may indicate that varied commuting hours expose flexible commuters to a range of crowding conditions.

Hypothesis 2 Testing

“Access to Real-Time Crowding Information (RTCI) will significantly influence commuters' decisions to travel on Metro Manila’s train lines, leading them to adjust their travel times or seek alternative modes of transportation.”

Association Between RTCI Awareness and Likelihood of Usage

Contingency Tables

| RTCI Awareness | Likelihood to Use RTCI |          | Total |
|----------------|------------------------|----------|-------|
|                | Likely                 | Unlikely |       |
| Not Aware      | 165                    | 22       | 187   |
| Aware          | 56                     | 9        | 65    |
| Total          | 221                    | 31       | 252   |

$\chi^2$  Tests

|          | Value | df | p     |
|----------|-------|----|-------|
| $\chi^2$ | 0.19  | 1  | 0.660 |
| N        | 252   |    |       |

Fig. 7. Chi-square test for respondents' RTCI awareness and their reported likelihood of usage

A chi-square test of independence was conducted among the survey responses to examine the association between awareness of RTCI and the likelihood of using it to plan their commutes. The contingency table displays the frequency of respondents classified by RTCI awareness (Aware vs. Not Aware) and their reported likelihood of using RTCI (Likely vs. Unlikely). Results from the chi-square test showed no significant association between awareness of RTCI and the likelihood of using it ( $\chi^2 = 0.19$  and  $p = 0.660$ ). This suggests that respondents' awareness of RTCI, even after being provided with an explanation and visual representation of how RTCI works, did not have a

statistically significant impact on their likelihood to consider using it for their commute.

Logistic Regression Analysis of RTCI Awareness and Likelihood to Use RTCI on Commuting Behavior Change

Model Fit Measures

| Model | Deviance | AIC    | BIC    | $R^2_{McF}$ |
|-------|----------|--------|--------|-------------|
| 1     | 106.82   | 136.82 | 189.76 | 0.28        |

Model Coefficients - Behavior Change

| Predictor                                     | Estimate | SE   | Z     | p      |
|-----------------------------------------------|----------|------|-------|--------|
| Intercept                                     | 0.16     | 0.73 | 0.22  | 0.826  |
| RTCI Awareness Binary                         | -0.48    | 0.63 | -0.77 | 0.443  |
| Age Range:                                    |          |      |       |        |
| 25-34 – 18-24                                 | -0.49    | 0.87 | -0.57 | 0.570  |
| 35-44 – 18-24                                 | 0.09     | 0.92 | 0.10  | 0.924  |
| 45-54 – 18-24                                 | -0.36    | 1.20 | -0.30 | 0.763  |
| Under 18 – 18-24                              | 0.02     | 1.35 | 0.01  | 0.991  |
| 55 and above – 18-24                          | 1.44     | 1.34 | 1.08  | 0.282  |
| Commuting Frequency:                          |          |      |       |        |
| 3-4 days a week – 5-7 days a week (Daily)     | 0.43     | 0.71 | 0.60  | 0.549  |
| Irregular – 5-7 days a week (Daily)           | -0.35    | 0.87 | -0.40 | 0.687  |
| 1-2 days a week – 5-7 days a week (Daily)     | -1.57    | 1.19 | -1.32 | 0.188  |
| A few times a month – 5-7 days a week (Daily) | 0.21     | 0.82 | 0.26  | 0.795  |
| Likelihood to Use RTCI (Likert):              |          |      |       |        |
| 2 – 1                                         | -0.88    | 0.84 | -1.05 | 0.292  |
| 3 – 1                                         | -1.65    | 0.78 | -2.12 | 0.034  |
| 4 – 1                                         | -3.08    | 0.87 | -3.56 | < .001 |
| 5 – 1                                         | -4.86    | 1.23 | -3.96 | < .001 |

Note. Estimates represent the log odds of "Behavior Change = Would not change" vs. "Behavior Change = Would change"

Fig. 8 Logistic regression analysis of RTCI awareness and likelihood to adjust commuting behavior

The logistic regression analysis, with a moderate model fit (McFadden's  $R^2$  of 0.28) revealed that respondents' awareness of RTCI, age range, and commuting frequency did not significantly influence their likelihood of adjusting their commuting behavior. However, a strong association was found between the reported likelihood of using RTCI and the willingness to change behavior. Specifically, respondents who rated their likelihood to use RTCI as 4 or 5 (Likely or Very Likely) on the Likert scale were significantly more likely to report a change in their commuting behavior. This suggests that the intention to use RTCI is a key factor in influencing commuters' decisions to adjust their commuting behavior.

Qualitative Analysis for Open-Ended Questions

To explore the experiences of commuters on Metro Manila’s train lines, we analyzed responses to the open-ended questions in our survey. Thematic analysis of the questions about overcrowding identified recurring themes reflecting physical, emotional, and practical challenges faced by respondents. Semantic analysis was conducted to gauge commuters' sentiments toward the potential implementation of RTCI.

| SUMMARY OF THEMES IDENTIFIED FROM COMMUTERS' EXPERIENCES |                                                                                                              |
|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Theme                                                    | Description                                                                                                  |
| Physical Discomfort                                      | Overcrowding leads to increased heat, lack of sufficient ventilation, and unwanted close physical contact.   |
| Stress and Anxiety                                       | Overcrowding induces fear, frustration, and mental fatigue.                                                  |
| Travel Time Delays                                       | Overcrowding results in long waits and missed schedules.                                                     |
| Safety Concerns                                          | Commuters voiced concerns about risks like being trapped between doors, falling, or physically being pushed. |

|                            |                                                                                                 |
|----------------------------|-------------------------------------------------------------------------------------------------|
| Peak Hour Chaos            | Rush hour intensifies overcrowding. There is a lack of organization in boarding and deboarding. |
| Infrastructure Issues      | Poor air conditioning, lack of seating, and disorganized stations worsen the experience.        |
| Adaptation and Resignation | Overcrowding is seen as unavoidable, with some commuters resigned to endure it.                 |

Table II summarizes the seven key themes identified from responses about overcrowding on Metro Manila's train lines. Physical discomfort, such as heat and lack of ventilation, and travel time delays caused by overcrowding were prominent concerns.

Sentiment Analysis of RTCI Implementation

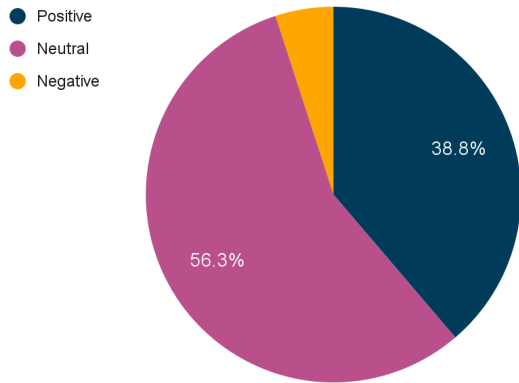


Fig. 9 Pie chart of the distribution of respondents' sentiments (positive, neutral, and negative) regarding RTCI implementation

The sentiment analysis regarding the potential implementation of RTCI revealed that most respondents held a neutral disposition. However, it is important to note that the question about RTCI was combined with a broader question about general improvements to Metro Manila's train systems. As a result, many of the neutral responses were not necessarily neutral toward RTCI itself but instead focused on providing constructive feedback and suggestions unrelated to RTCI, such as addressing systemic issues in the public transportation system.

Despite this, the analysis revealed that 38.8% of respondents to the open-ended questions explicitly expressed optimism and a positive outlook regarding the implementation of RTCI. Neutral feedback, which accounted for 56.3% of responses, primarily focused on suggestions not necessarily related to RTCI, such as infrastructure and management changes. Meanwhile, 5% of respondents voiced concerns about the accuracy of the data and systemic issues like inadequate train capacity and unreliable schedules—challenges that RTCI alone cannot address.

## B. Discussion of Results

### *Association Between Platform Overcrowding and Longer Waiting Times to Higher Levels of Discomfort*

For both non-peak and peak commuters, longer waiting times were significantly associated with decreased comfort levels, supporting the notion that extended waiting times contribute to commuter discomfort. However, the frequency of overcrowding did not have a statistically significant impact on comfort levels in either group. This suggests that

while crowding is a common experience, waiting times have a stronger effect on commuter comfort than the frequency of crowding. For flexible commuters, neither waiting time nor overcrowding frequency significantly impacted comfort levels. Instead, comfort for this group was more strongly associated with positive experiences during both peak and non-peak hours. This implies that positive commuting experiences during varied commuting conditions may serve as a buffer against the discomfort caused by waiting or crowding.

Overall, the findings highlight that waiting times are a key factor influencing discomfort in fixed-schedule commuters, while the impact of overcrowding frequency is less pronounced. These results partially support our hypothesis that longer waiting times, most often caused by overcrowding, lead to lower comfort levels.

### *Association Between RTCI Awareness and Likelihood of Usage*

The lack of a significant association between RTCI awareness and the likelihood of RTCI usage indicates that awareness alone is not a strong predictor of commuter interest in using RTCI for planning their commute. Given that respondents were introduced to RTCI through a brief explanation and visual aid in the survey, "awareness" in this context may reflect an immediate understanding of the topic rather than prior knowledge or interest. This outcome suggests that awareness, even when accompanied by information about RTCI's purpose and functionality, may not be sufficient to drive behavioral change among commuters. For RTCI to influence commuting habits, other factors—such as trust in the data's accuracy, perceived relevance to personal schedules, or other tangible benefits such as application features—may need to be considered.

### *Qualitative Analysis for Overcrowding and RTCI Implementation*

The thematic analysis of the challenges faced by commuters revealed several recurring themes related to overcrowding. Respondents frequently mentioned physical discomfort, especially during peak hours. This discomfort was closely linked to stress and anxiety, as many commuters associated overcrowding with mental strain due to increased general frustration and fears of theft. Travel time delays were another significant issue, with overcrowding forcing commuters to wait for multiple trains, often resulting in tardiness for work or arriving home exhausted. Safety was also a major concern. Respondents reported risks such as being trapped between train doors or being pushed aggressively during peak overcrowding periods. Infrastructure issues like inadequate air conditioning, insufficient seating, and poorly designed stations further worsened overcrowding's negative impacts. Lastly, some commuters expressed a sense of resignation. They accepted overcrowding as an unavoidable part of their daily commute, choosing to endure unfavorable conditions. These findings highlight the need for innovative solutions such as an RTCI system that can help redistribute passengers and lessen overcrowding.

Overall, the analysis highlights an interest in RTCI and its potential benefits, but it also suggests that its



implementation may need to be accompanied by systemic reforms to address broader commuting challenges effectively. Our findings align with existing literature emphasizing the importance of real-time information systems in enhancing commuter experiences.

### C. Theoretical Contribution and Practical Implications

This study contributes to the theoretical understanding of commuter behavior in urban transit systems by highlighting how RTCI can potentially address overcrowding on Metro Manila's train lines. The findings expand existing frameworks on the adoption of technologies such as RTCI in public transportation by demonstrating how it can significantly affect passenger route choices [15] and influence commuter decision-making. The use of real-time crowding data in decision-making frameworks aligns with previous studies that highlight the predictive power of smart card data for understanding and managing crowding dynamics in metro systems [16]. Their study underscores the importance of leveraging such data in optimizing passenger flow and creating a more responsive transport system.

RTCI systems can be designed to work with a variety of sensing technologies [17] to provide live updates on overcrowding and waiting times. For example, crowding data can be inferred from Automatic Fare Collection (AFC) systems [18] or through mobile phone accelerometers [19]. In line with this, predictive models can be utilized to deliver crowding information sufficiently early to influence travel routes [20]. These existing technologies can be used to develop an RTCI system that can provide valuable decision-making data to consumers.

The practical implications of this research are significant for policymakers, transit authorities, and most notably developers of RTCI systems. The findings can support policymakers and transit authorities in justifying investments in RTCI as part of broader urban transport initiatives. They also provide relevant data for RTCI developers to enhance or tailor their systems to meet specific commuter needs. Beyond Metro Manila, the findings present a framework through which similar systems could be adopted.

Finally, this study reinforces the role of information systems in managing urban transit by contributing to the development of smarter, more efficient public transportation systems.

### D. Limitations

While this study provides valuable insights into commuter experiences with overcrowding on Metro Manila's train lines and the potential impact of RTCI, several limitations must be acknowledged.

First, the sample size of 252 respondents may not have fully captured the diverse commuter population of Metro Manila. The use of convenience sampling resulted in a sample size that includes mostly young, more technologically-inclined adults who are more likely to engage in online surveys. This limited sample size restricts the generalizability of the findings, and the results should be interpreted as indicative rather than representative of the entire commuter base.

From a technological perspective, this study assumed the hypothetical availability of RTCI and did not delve into the complexities of implementing such a system. Therefore, the study did not account for technical limitations or infrastructure challenges that may affect the successful deployment of RTCI in Metro Manila's train systems.

Another limitation lies in the temporal scope of data collection, which was confined to a five-week period. This timeframe may not have captured variations in commuter behavior that can occur across different seasons or in response to long-term changes in train operations and infrastructure.

Finally, while the study explores commuters' experiences and potential responses to RTCI, it did not address other possible factors that can influence commuting decisions, such as fare prices, alternative transportation options, or transportation policies. Perceptions of crowding differ based on various factors such as mode of transport, study area, and data type [21]. These factors are critical for understanding the broader context in which RTCI systems can operate most efficiently. However, they were outside the scope of this study.

## IV. CONCLUSION

This study explored the experiences and perceptions of commuters regarding overcrowding on Metro Manila's train lines (LRT-1, LRT-2, and MRT-3) and explored the potential impact of Real-Time Crowding Information (RTCI) on commuter behavior. The findings indicate that overcrowding remains a significant issue, with commuters experiencing discomfort, longer waiting times, and increased stress, particularly during peak hours. The introduction of RTCI was met with positive interest, with many commuters expressing optimism about its potential to alleviate some of their commuting challenges.

However, the study found that awareness of RTCI alone did not strongly influence commuters' likelihood of using the system. While RTCI holds potential, its impact on commuter behavior may depend on additional factors, such as trust in the accuracy of data, the perceived relevance to personal schedules, and the inclusion of more tangible features and benefits. These concerns highlight the necessity of developing a user-friendly and accessible mobile application that is relevant to commuters' needs. An RTCI mobile application can bridge the gap between awareness and active usage by enabling commuters to make informed decisions based on crowding levels, waiting times, and train schedules. This information could drive behavioral change, making RTCI a more integral part of commuters' travel decision-making process.

In conclusion, this research contributes to the ongoing dialogue around public transportation in Metro Manila and provides valuable insights for policymakers, transit authorities, and developers of RTCI systems. Future research should explore the technical aspects of RTCI implementation and examine how it can be optimized to address commuter needs. This study serves as a foundation for further investigations into the potential of information systems to improve urban public transport and enhance the overall commuting experience.

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